

Assignment 1

Due: 26th April 2026

Assignment Instructions

This assignment requires you to perform a series of controlled experiments on a unique physical object of your choosing (e.g., a specific mechanical tool, a circuit board, or a piece of architectural hardware).

Instructions

1. Fourier Transform

- Capture an image of your object. Intentionally introduce a specific directional noise (e.g., motion blur in one direction or a periodic pattern).
- Perform a Fourier Transform. You must identify the specific "noise spikes" in the spectrum.

Note: You must submit a screenshot of the Fourier spectrum *before and after* you manually masked the specific frequency coordinates. Explain why a vertical bright line appeared in your spectrum and how your Low-Pass filter specifically targeted the noise without destroying the object's primary edges. (5mks)

2. Canny Edge Detection

- Implement Canny Edge Detection on the object image
- You must produce two versions of the edge map:
 - **Under-segmented:** Where "Strong Edges" are lost.
 - **Over-segmented:** Where "Candidate Edges" create a spiderweb of noise.

Note: Document the exact grayscale values at which your object's specific features disappeared. (5mks)

3. The Hough Transform

- Detect the primary lines or circles of your object using a Hough Transform.

4. Harris Corner

- Detect corners on your object.
- Physically rotate your object by exactly 45 degrees and re-run the Harris detector.
- Compare the "Eigenvalue" response of the same corner in both orientations. Explain mathematically why the Harris detector remained "invariant" to this rotation while a simple edge detector would have changed its gradient direction. (5mks)

5. Haar Cascade

- Run a Haar Cascade to localize a feature.
- Manually define a single 3x3 Sobel kernel and time its execution across the image vs. a 3-stage Haar Cascade. (5mks)

Submission Requirements

- **Jupyter Notebook:** Must contain the outputs of the specific experiments mentioned above.
- **Technical PDF:** Must contain the results of the experiments which should be well cited. (Not more than 3 pages) (10mks)